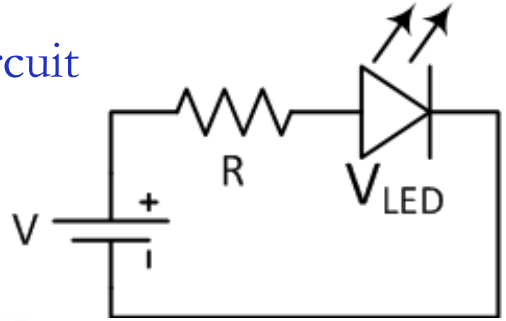
Light Emitting Diodes

The following subjects will be covered

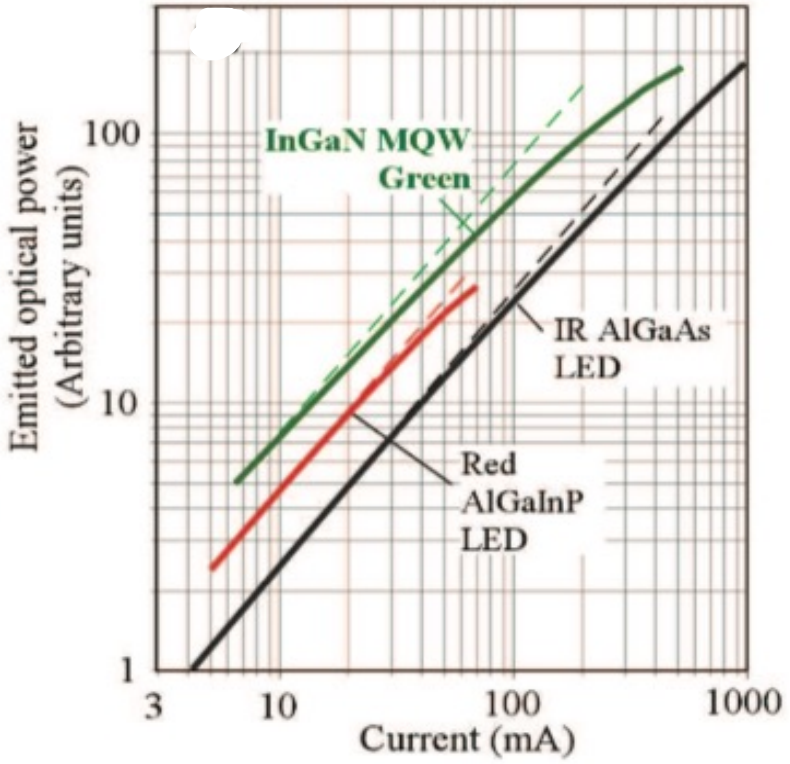
- The PN Junction Band Diagram.
- LED Basic Design and Operation.
- LED Efficiency.
- Improved LED Designs.
- Basic LED Types.
- LED Characterization.
- LED Materials.

LED Characterization
Output Optical Power

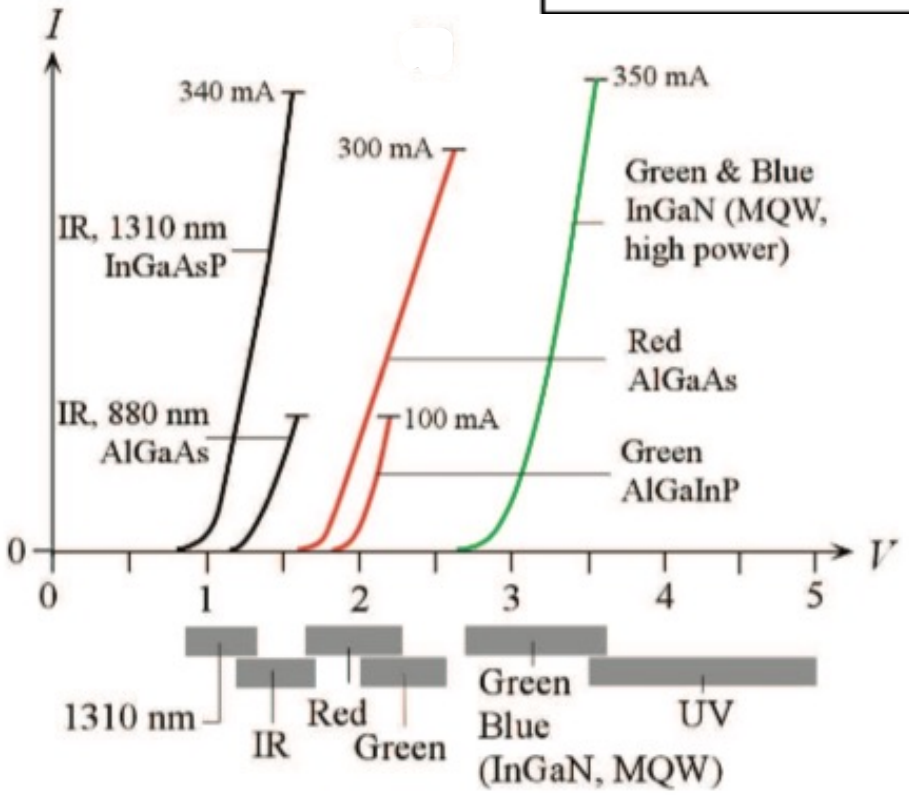
Circuit



PI Curve $P_o = \hbar\omega\, \eta_{ex}\, \frac{i}{q}$



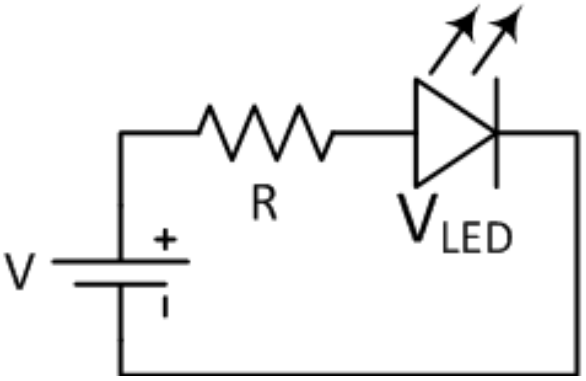
IV Curve

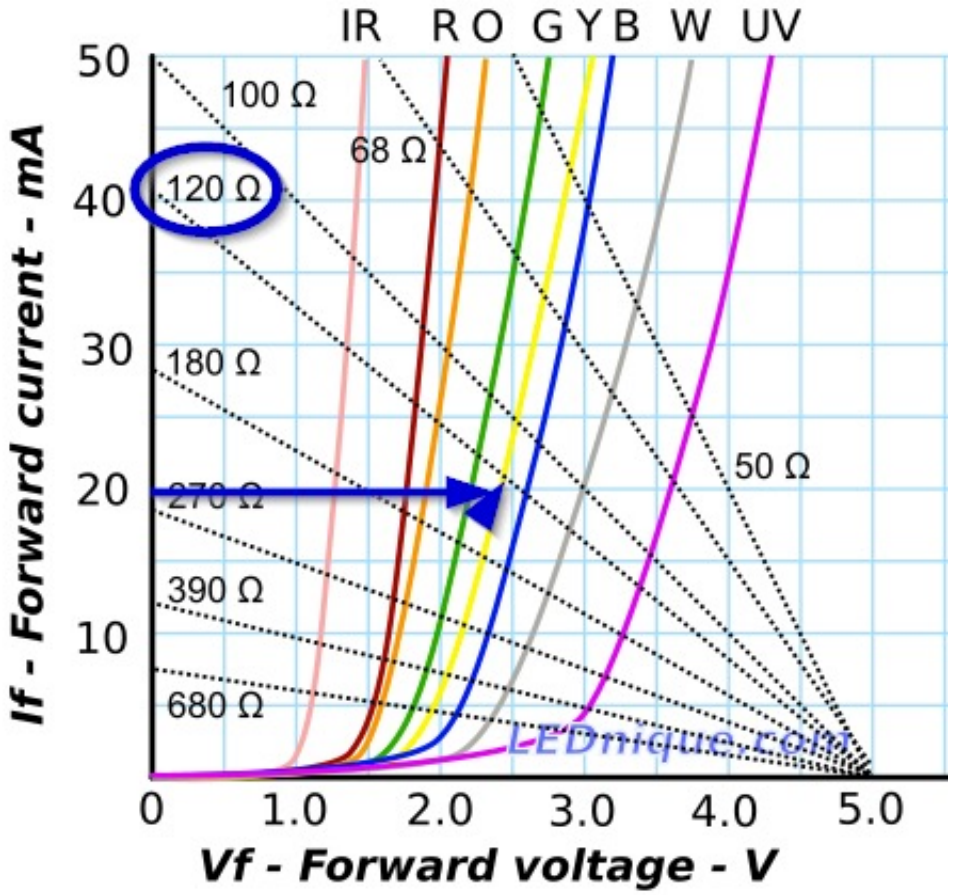


- Carrier leakage
- Joule heating

LED Characterization
Output Optical Power

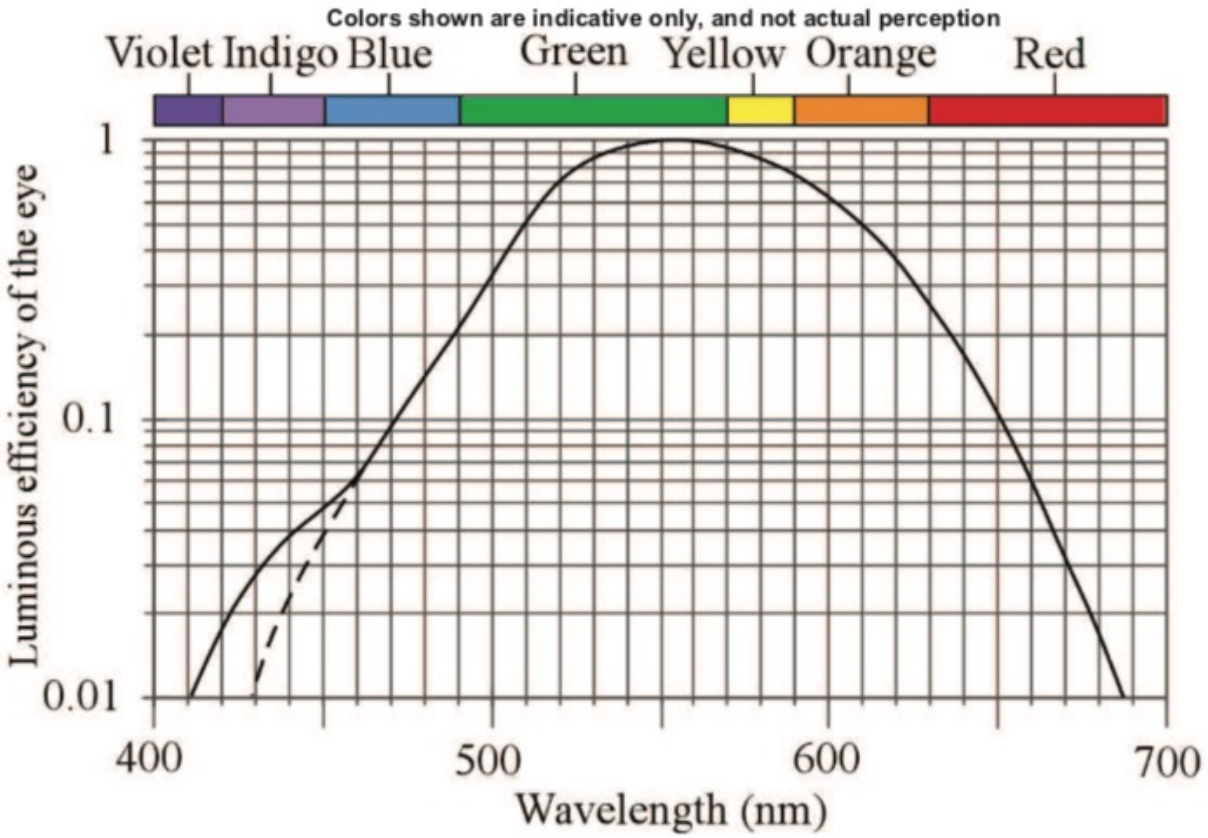
Example: Find the value of the series resistance for a yellow LED biased at 20 mA. Take V=5 V.





LED Characterization
Output Optical Power

The human eye optical spectrum: Luminous efficiency.



Example: What is the luminous flux of a red LED emitting at 650 nm with output power of 10 mW?

Luminous efficiency at 650 nm is 0.1.

Luminous flux is given by

$$Lum = \eta_{LE} P_o (633\ lm/W)$$

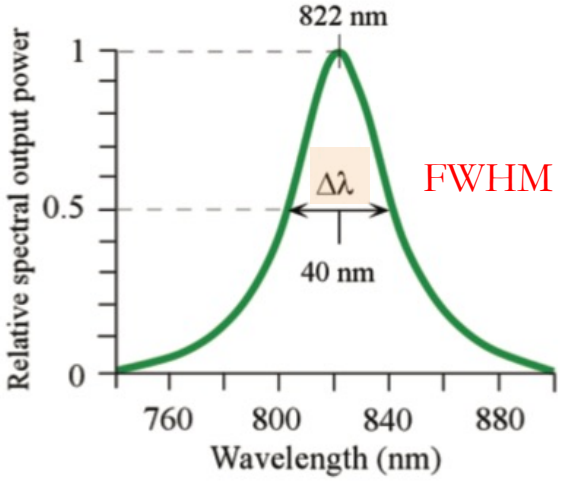
$$Lum = 0.63\ \text{lm}$$

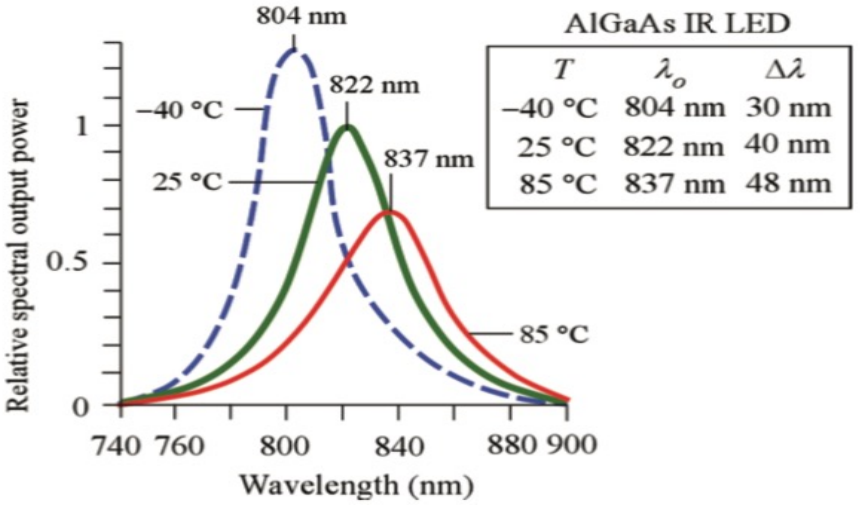
Repeat for green at 532 nm.

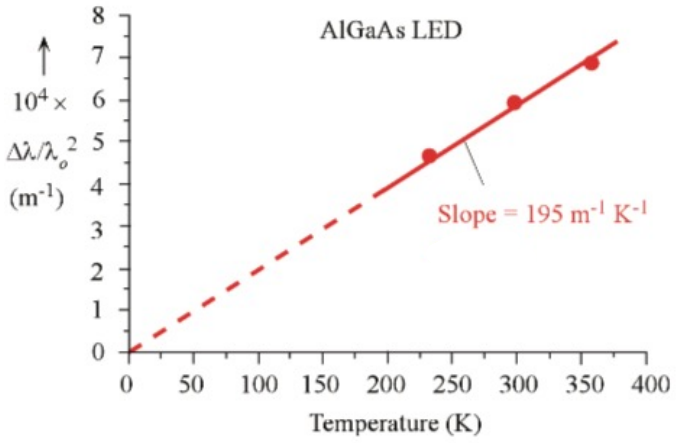
➡ $Lum = 5.51\ \text{lm}$

LED Characterization
Spectral Distribution

Spectral Distribution of an LED is described by peak wavelength λ_o and spectral width $\Delta\lambda$.







$$E_G = E_{Go} - \frac{AT^2}{(B+T)}$$ Varshni Equation

- Red-shift with temperature is due to decrease in bandgap.
- Increase of spectral width with temperature is due to electron-hole distribution in the conduction/valence bands.

LED Characterization
Spectral Distribution

We can express the spectral width $\Delta\lambda$ in terms of change in photon energy $\Delta\hbar\omega$.

$$\lambda = \frac{hc}{E}$$

$$\frac{\Delta\lambda}{\Delta E} = hc\left(\frac{-1}{E^2}\right)$$

$$\Delta\lambda = -hc\left(\frac{\Delta E}{E^2}\right)$$

$$\Delta\lambda = -\left(\frac{\lambda_o^2}{hc}\right)\Delta E$$

$$\frac{\Delta\lambda}{\lambda_o^2} = -\frac{\Delta E}{hc}$$

$$\Delta E \approx 2.5 - 3\,k_B T$$

Example: Find the spectral width for a GaAsP LED emitting at 655 nm at 300 K. Take $\Delta E = 3\,k_B T$.

$$\Delta\lambda = \left(\frac{(655\times10^{-9})^2}{6.626\times10^{-34}\times3\times10^{8}}\right)(3\times1.38\times10^{-23}\times300)$$

$$\Delta\lambda = 26.8\,nm$$

Example: Find the shift in the emitted wavelength of a GaAs LED per 1 °C change at room temperature. E_{Go}=1.519 eV, A= 5.41×10^{-4} eV/K, B=204 K.

$$\frac{\Delta E_G}{\Delta T} = -\left(\frac{E_{Go} - E_G}{T}\right)\left(\frac{T + 2B}{T + B}\right)$$

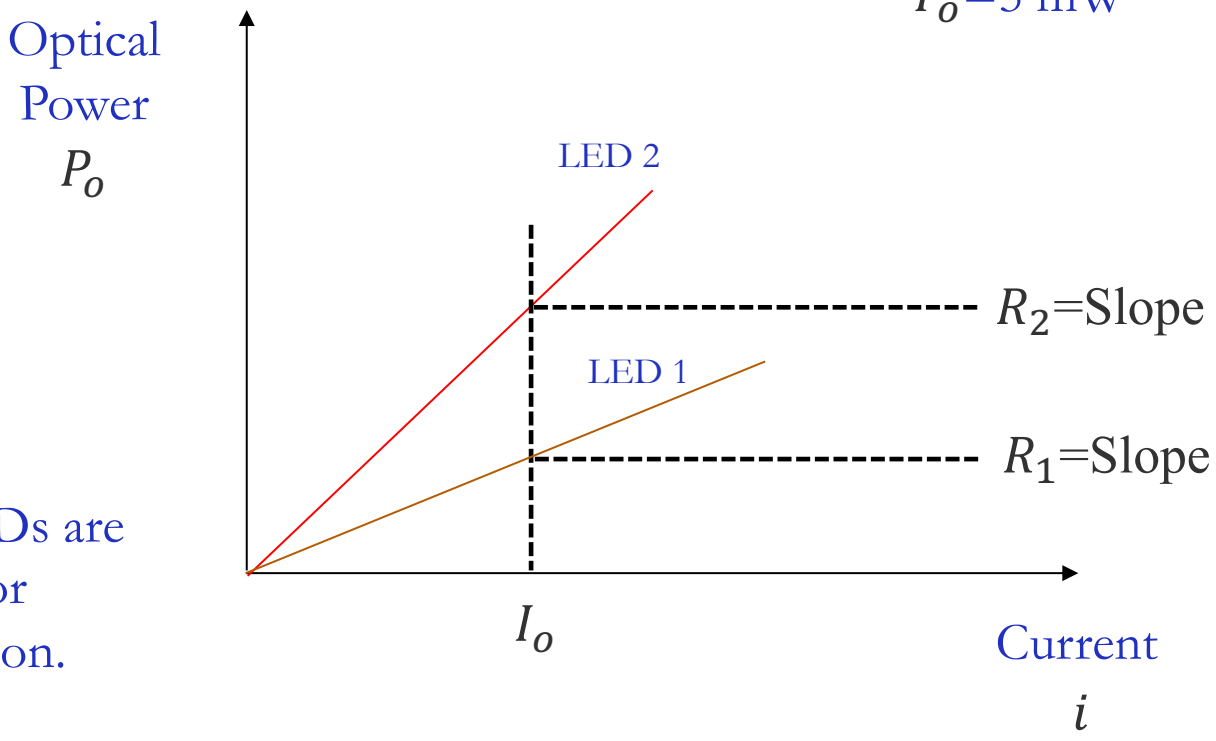
LED Characterization
Responsivity

LED Responsivity is defined as,

$$R = \frac{P_o}{i} = \hbar\omega\,\eta_{ex}$$

Example: Consider an LED with R=50 μW/mA, find the output power for I_o=100 mA.

P_o=5 mW



High R LEDs are used for modulation.

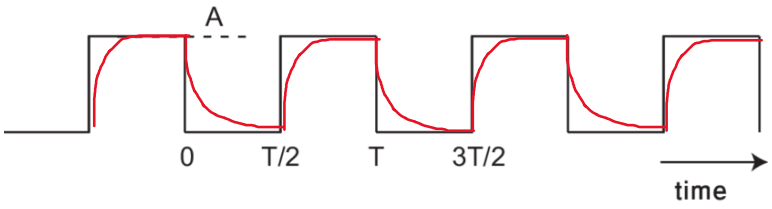
LED Characterization
Modulation Bandwidth

The ability of the LED to follow the speed of transitions of the AC signal.

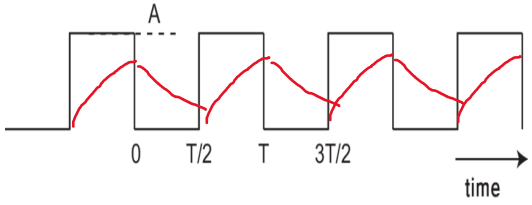
Main factors

1) Carrier recombination time (optical).
2) Circuit RC delay (electrical).

Low frequency



High frequency

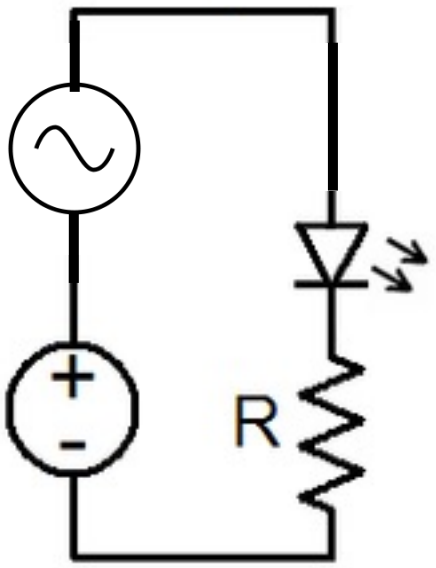


Good communication LEDs have 300-700 MHz bandwidth

$\rightarrow i = I_{DC} + i_{ac}$

AC signal

DC Bias



Optical cut-off frequency

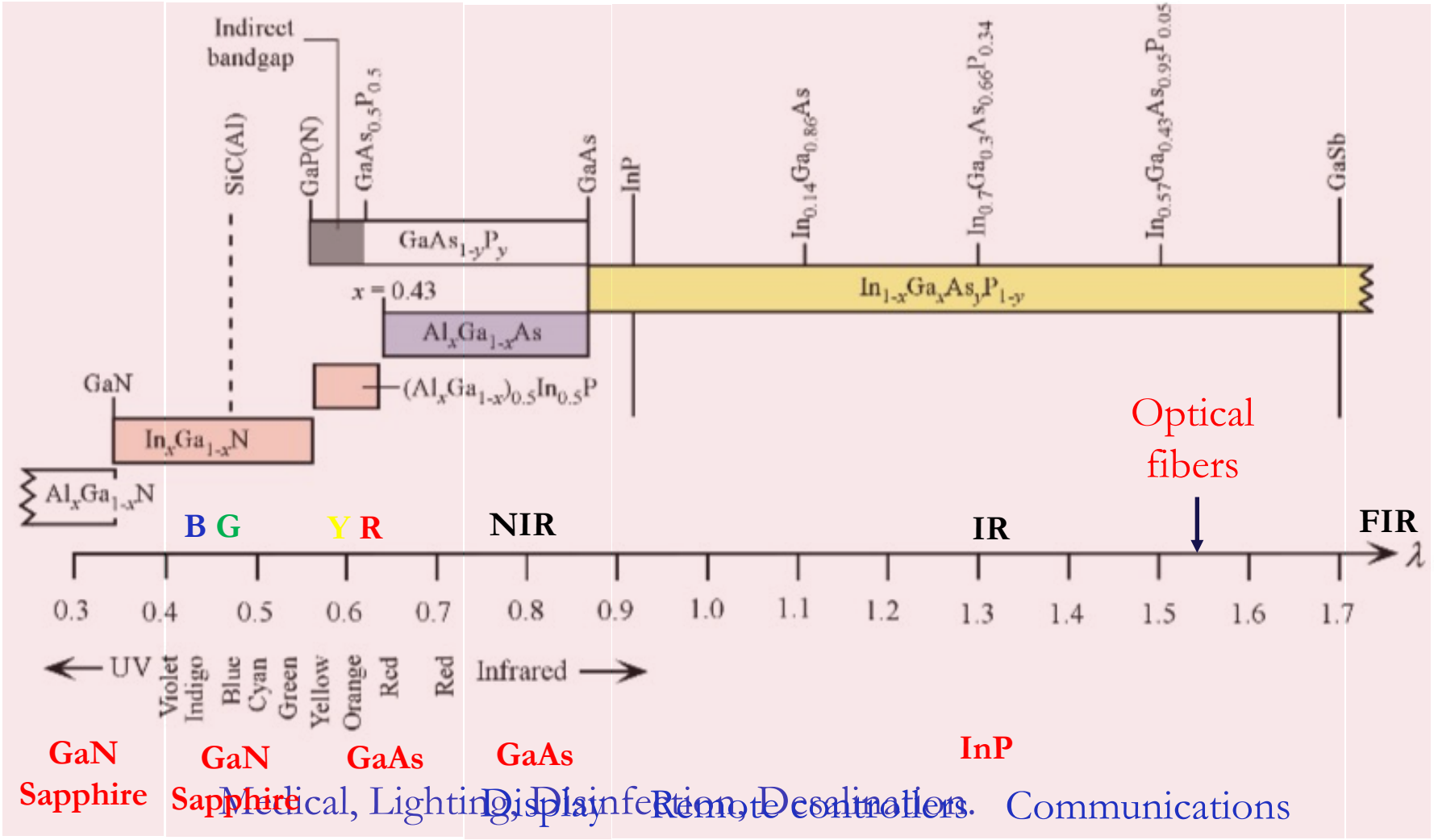
$$f_c = \frac{\sqrt{3}}{2\pi\tau}$$

Electrical cut-off frequency

$$f_c = \frac{1}{2\pi RC}$$

LED Materials

Materials and Applications.



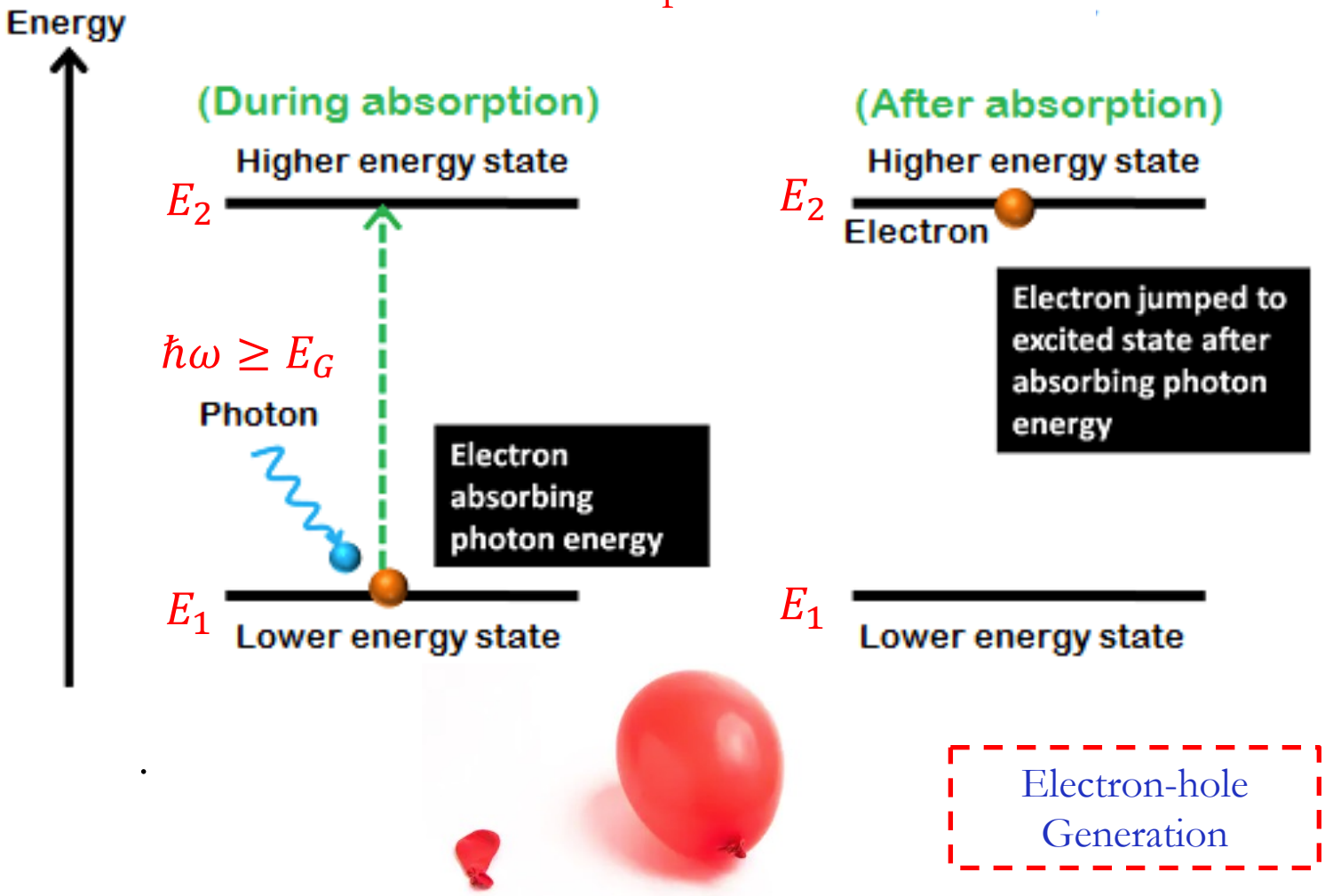
LASERS

The following subjects will be covered

- Stimulated Emission.
- Laser System Basic Components.
- The Fabry-Perot Cavity.
- Semiconductor Lasers Fundamentals.

Photon-matter Interactions

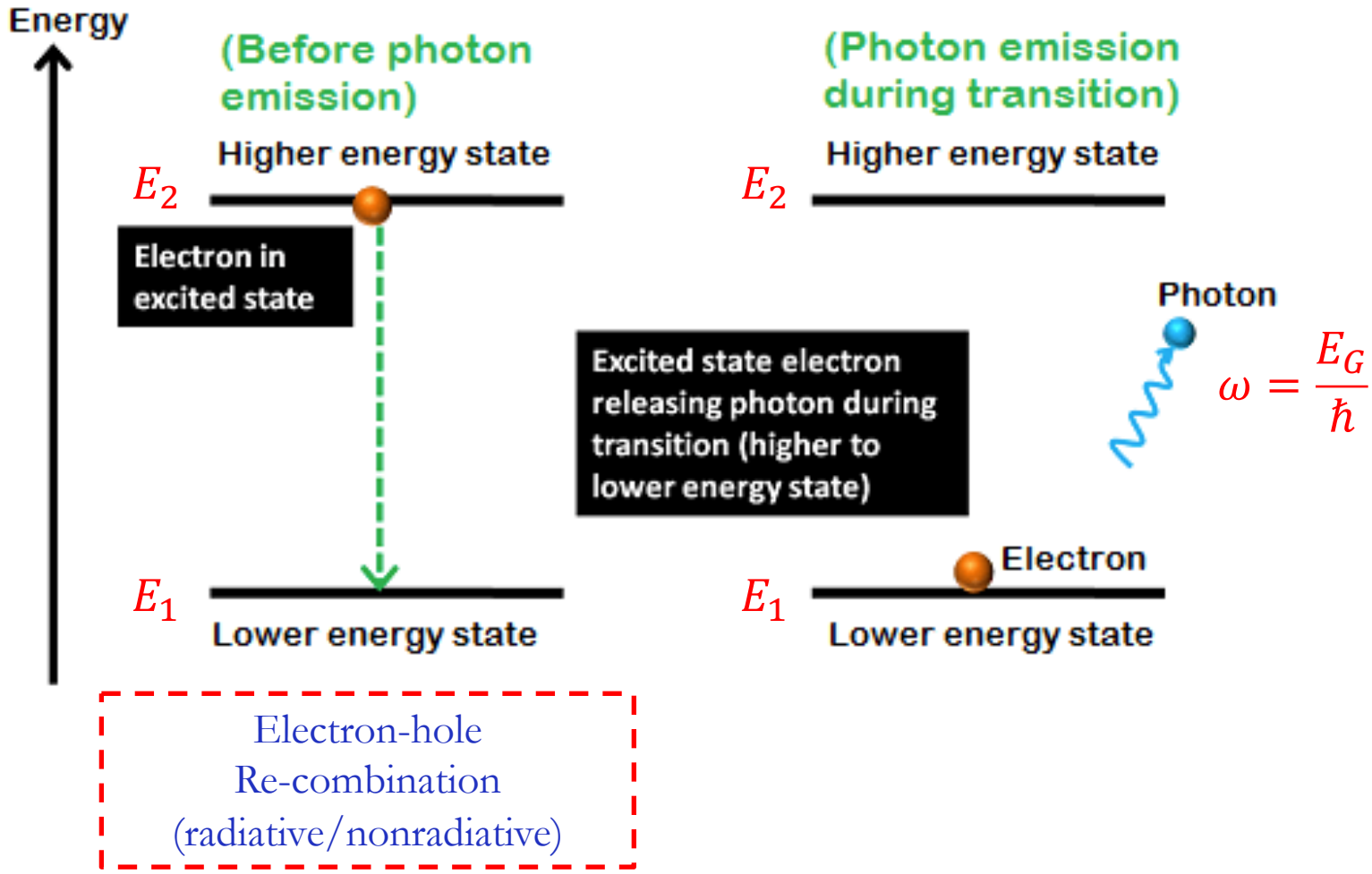
Absorption



Electron-hole Generation

Photon-matter Interactions

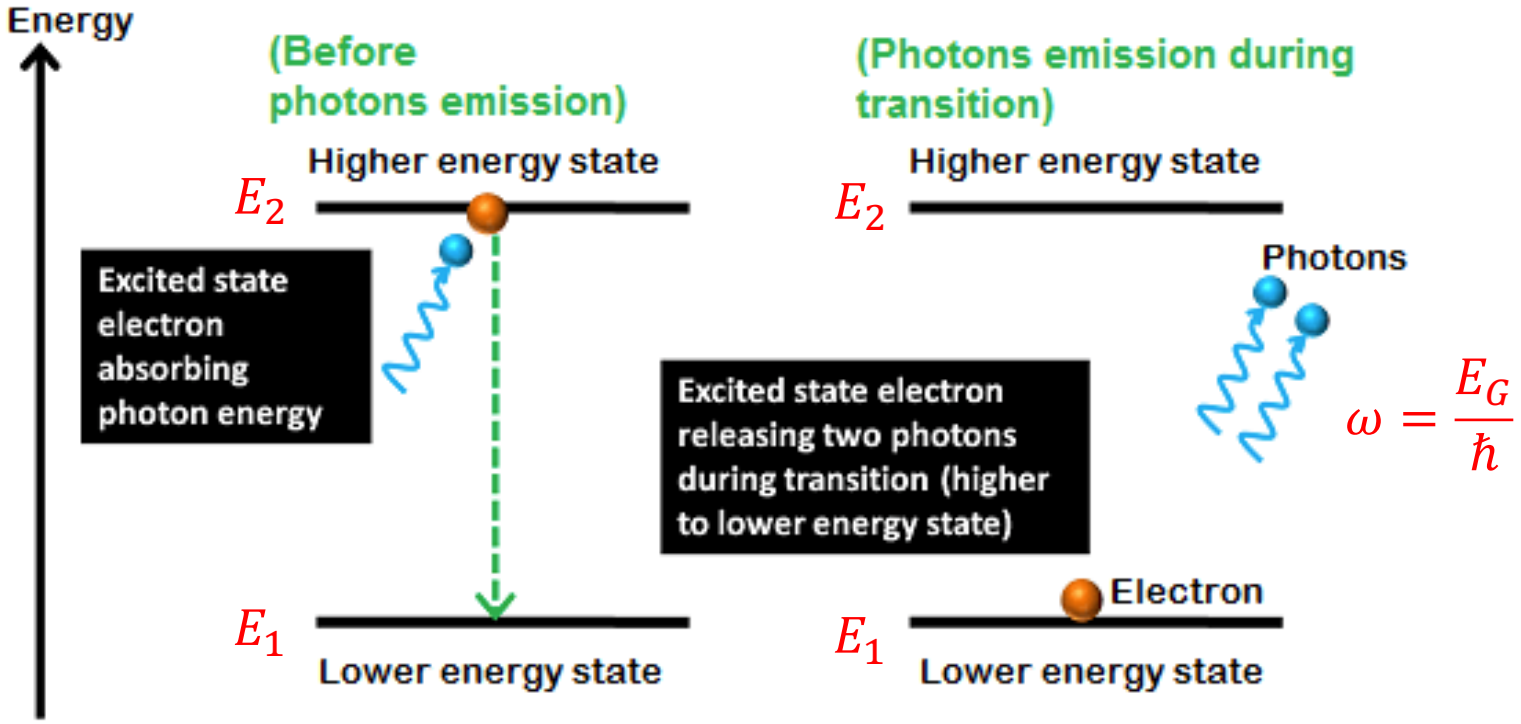
Spontaneous Emission



Electron-hole Re-combination (radiative/nonradiative)

Photon-matter Interactions

Stimulated Emission

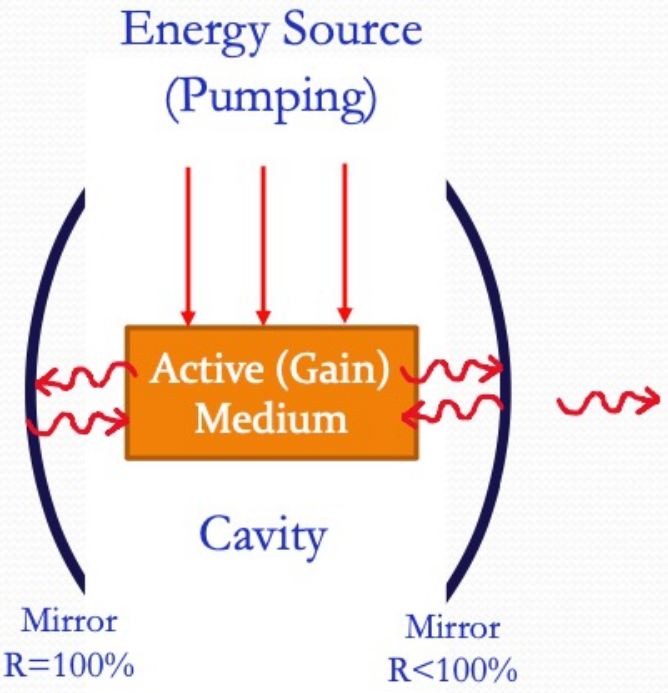


Electron-hole
Re-combination
(radiative/nonradiative)

LASER Components

Any laser system is composed of the following three components.

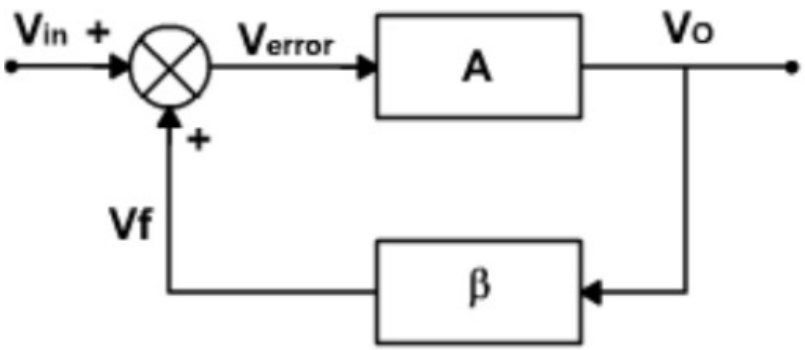
- Active (gain) medium: A medium that provides the necessary condition for the sustained generation of light. Multi energy level system with "population inversion".
- Energy source: An external source capable of producing required atomic excitations within the gain medium. Usually optical or electrical.

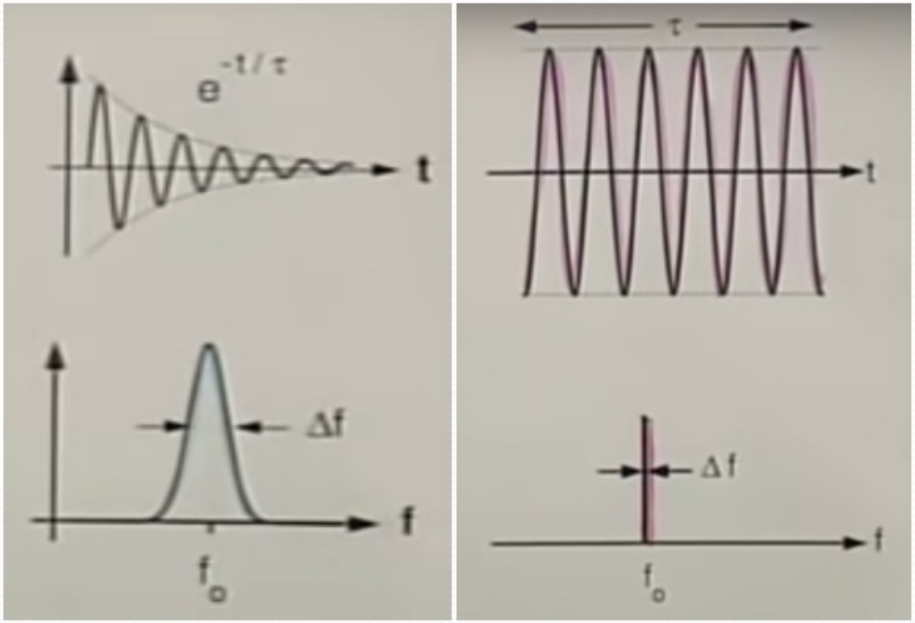


- Cavity: A mirror system (or equivalent) is used to reflect the incoming light in a constructive interference manner. One side is not perfectly reflecting to allow laser output light.

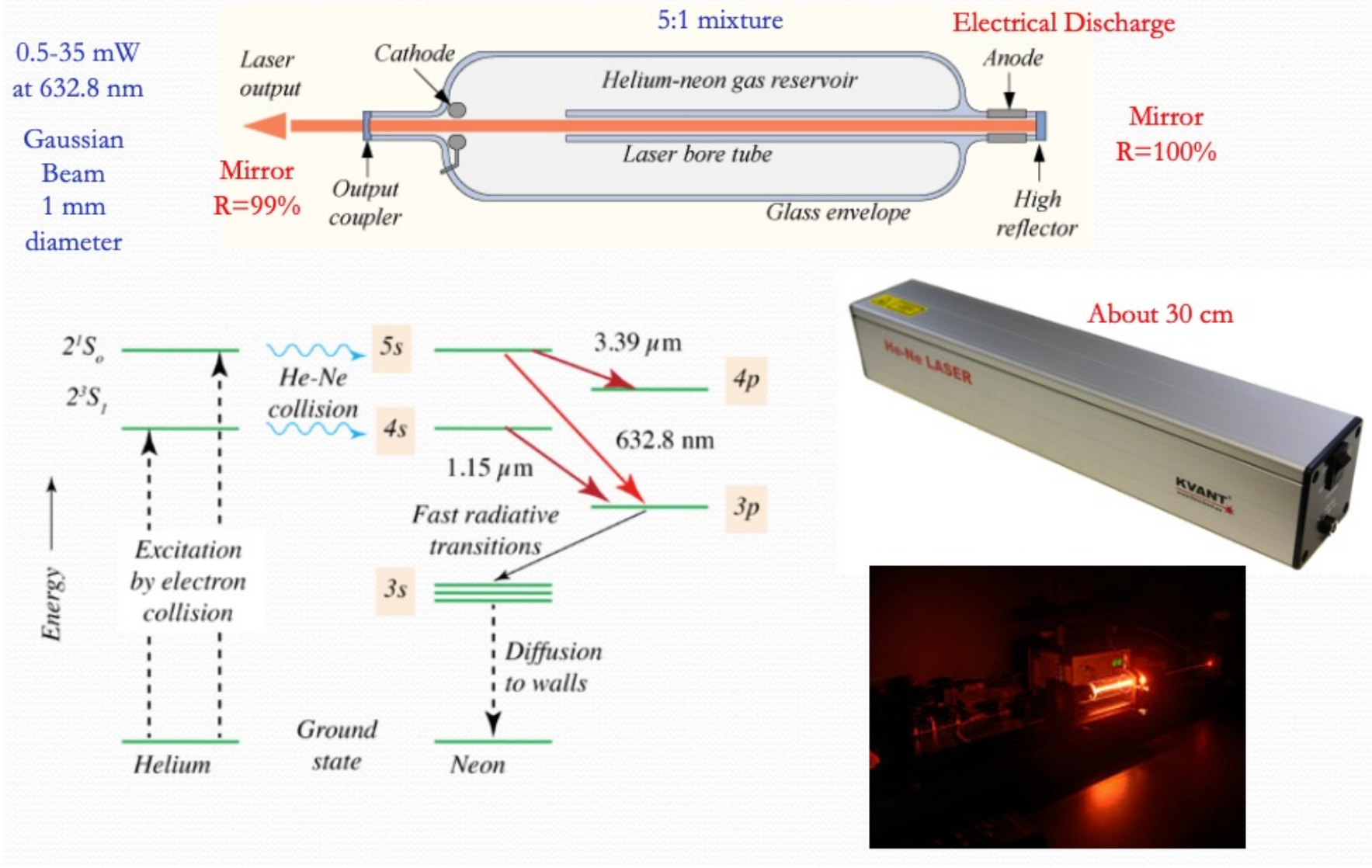
The Electronic Oscillator Analogy

The main problem is that these oscillations decay to zero as soon as the energy in the circuit is consumed, they are not sustained. Circuit loss should be compensated.



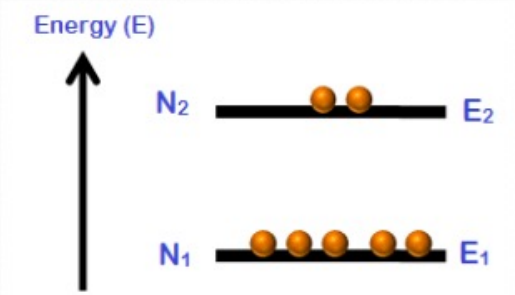


Example: He-Ne Laser

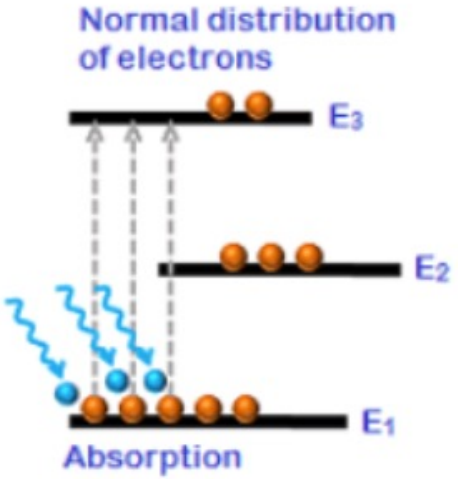


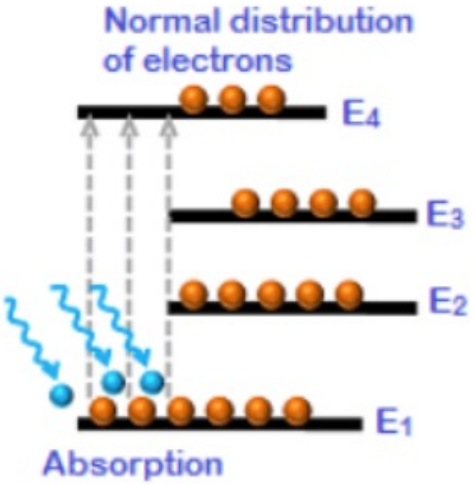
Population Inversion

Two-level Material Systems



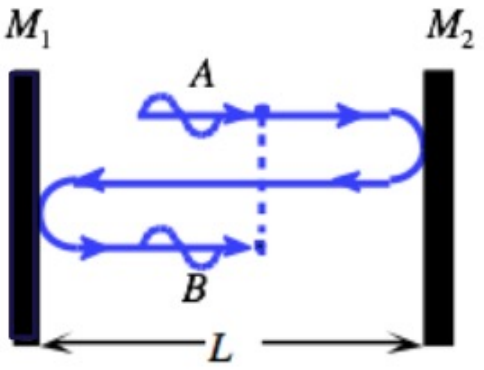
Three and Four-level Material Systems

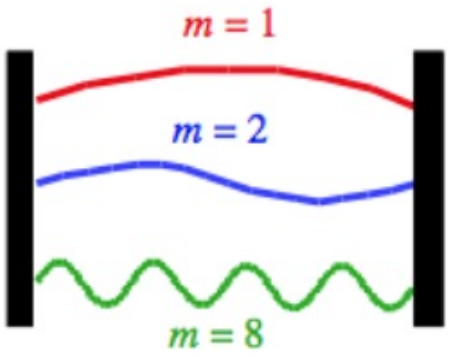




The Fabry-Perot Cavity

Two flat mirrors face-to-face, separation L.





$$L = m\,\frac{\lambda}{2} \quad (m=1, 2, 3, \ldots)$$

$$\lambda_m = \frac{2L}{m} \quad (m=1, 2, 3, \ldots)$$

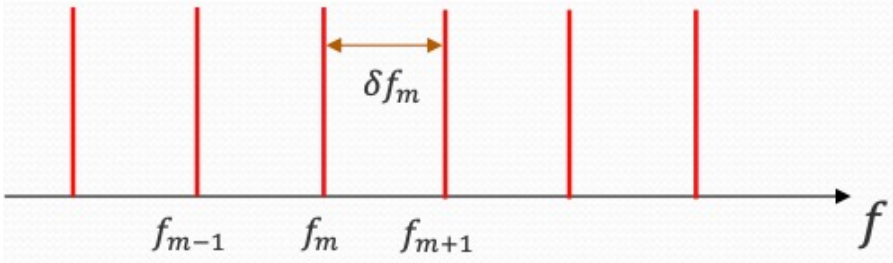
$\lambda_1, \lambda_2, \lambda_3, \ldots$ cavity modes

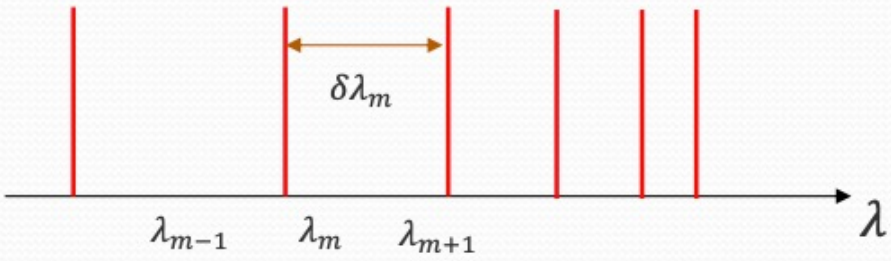
$$f_m = m\,\frac{c}{2L} \quad (m=1, 2, 3, \ldots)$$

$$\delta f_m = f_{m+1} - f_m = \frac{c}{2L}$$

(free spectral range)

$$\delta\lambda_m = \frac{2L}{m^2}$$





Example. In a GaInAsP (n =3.5) cavity (L=400 μm), what is the mode index at 1500 nm.

$$\lambda_m = \frac{2L}{m} \quad \Rightarrow \quad m = \frac{2L}{(\lambda_m/n)} = 1864$$

The Fabry-Perot Cavity

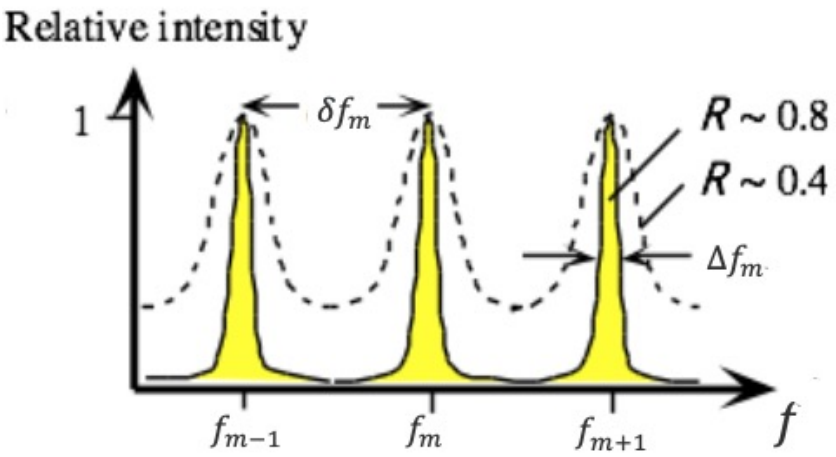
Field intensity inside the cavity.

$$E_{cavity} = \frac{A}{1 - Re^{-j2kL}} \quad \Rightarrow \quad I_{cavity} = |E|^2 = \frac{I_o}{(1-R)^2 + 4R\sin^2 kL}$$

R=100% gives $I_{cavity,max} = \infty$ at f_m

R<100% gives $I_{cavity,max} = \frac{I_o}{(1-R)^2}$ at f_m

Spectral width $\Delta f_m = \delta f_m \frac{(1-R)}{\pi\sqrt{R}}$



Example. Consider an air FP cavity of length 100 μm and R=0.9. Find the cavity mode near 900 nm.

$$\lambda_m = \frac{2L}{m} \quad \Rightarrow \quad m = \frac{2L}{\lambda_m} = 222.22$$

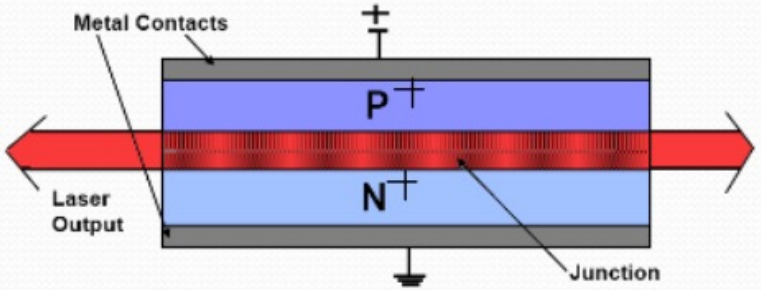
$\lambda_m = 900.0\ nm$

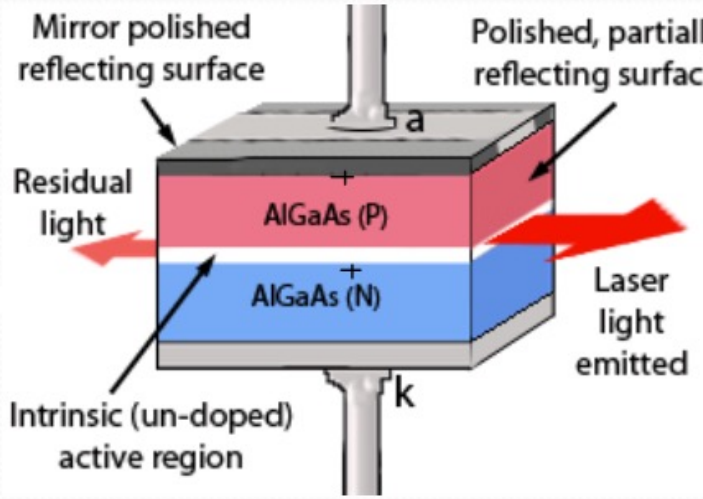
$$\delta f_m = \frac{c}{2L} = 1.5\times10^{12}\ \text{Hz}$$

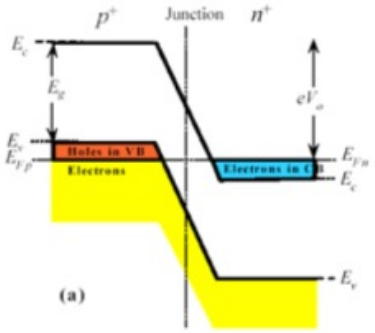
$$\Delta f_m = \delta f_m \frac{(1-R)}{\pi\sqrt{R}} = 5.03\times10^{10}\ \text{Hz}$$

Semiconductor Lasers

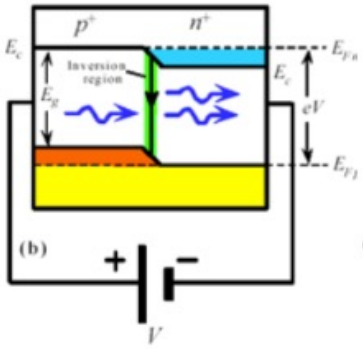
The basic homojunction structure is given below. The n and p sides are degenerately doped.



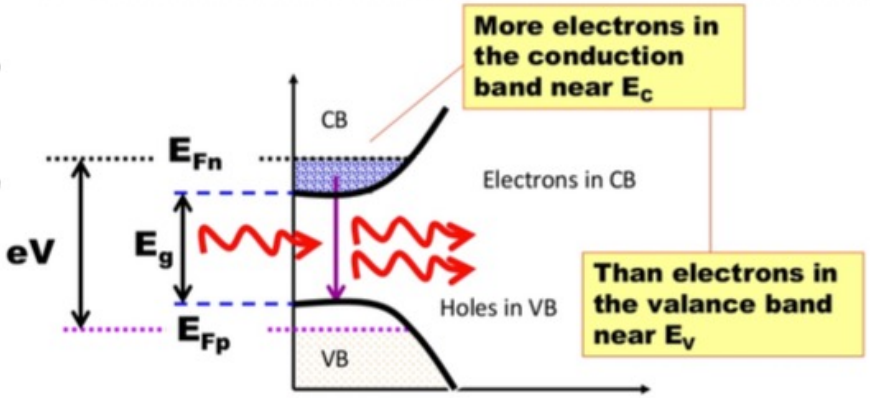




(a) p-n junction without bias



(b) p-n junction after forward bias



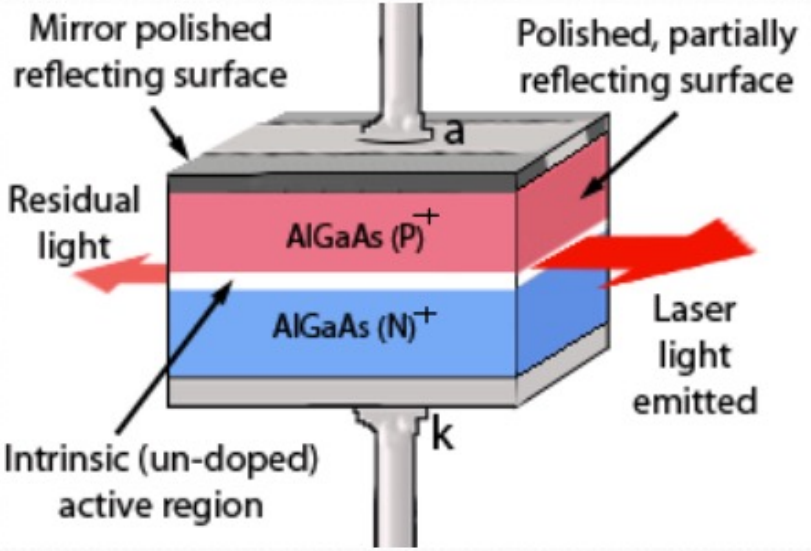
Semiconductor Lasers

Requirements for a laser diode:

1) Degenerate doping of the n and p sides (n^+ and p^+). The Fermi level of the n type is inside the conduction band and the Fermi level of the p type is inside the valence band. Population inversion is generated.

2) Forward bias the diode above threshold current to trigger stimulated emission.

3) Polished device walls for better reflection and hence create a cavity.



Heterostructure lasers

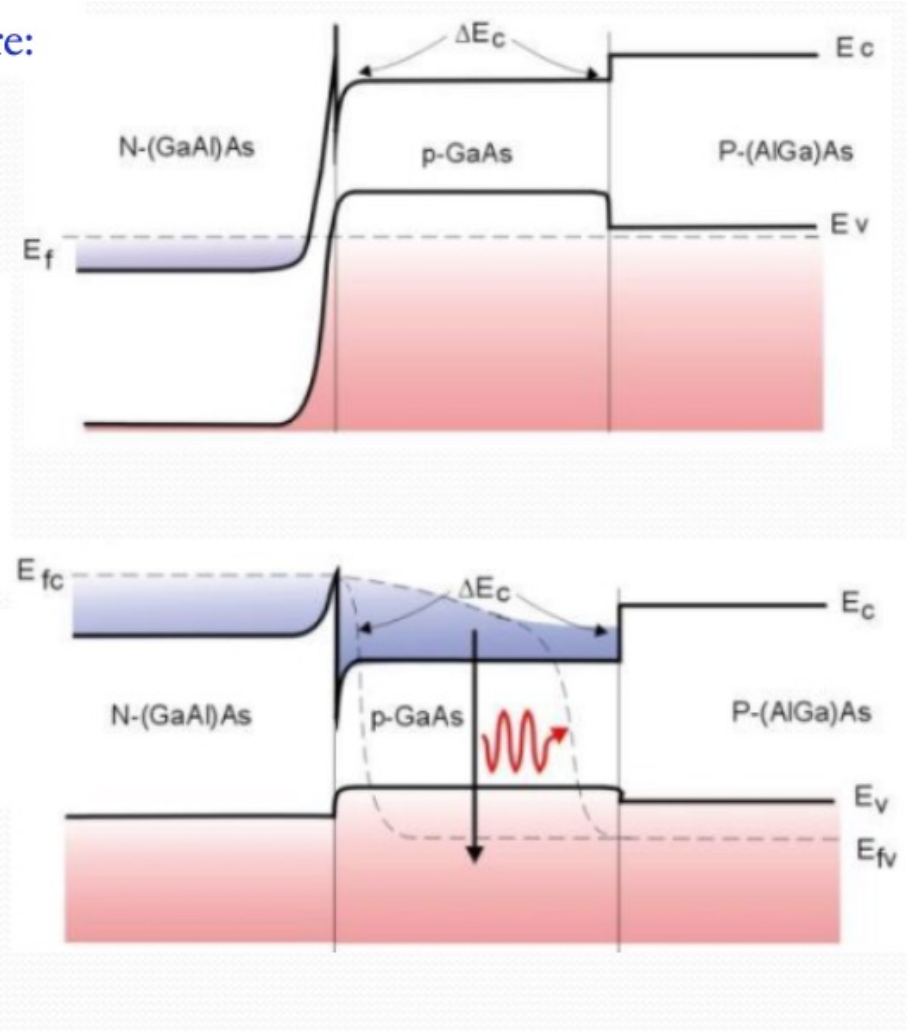
Advantages over homojunction structure:

1) Carrier confinement. Minimize leakage current.

2) Optical confinement. Guide the light towards the cavity walls (vertical confinement). The horizontal confinement is provided by the current path.

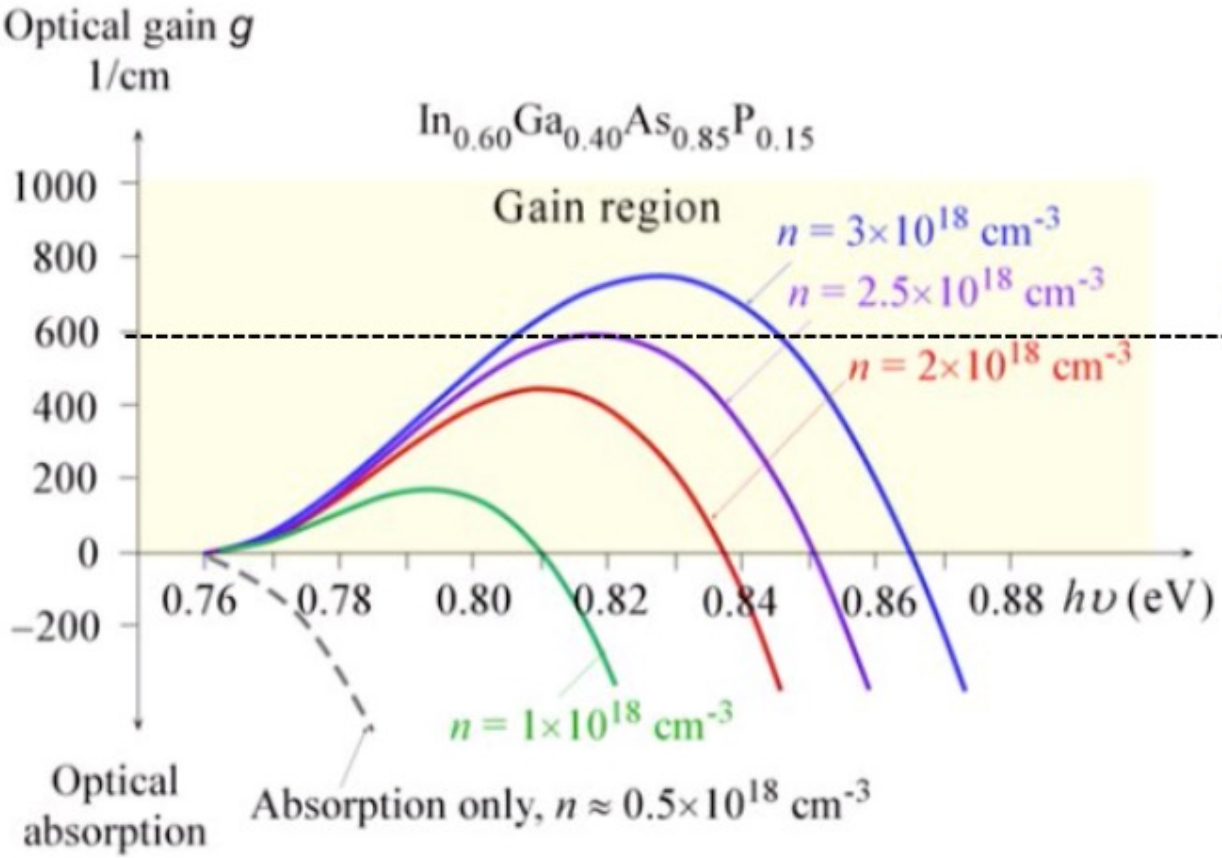
3) Minimum re-absorption. The large bandgap material does not allow band transition.

4) Lower threshold current (Nobel Prize 2000 for milliamp laser operation).



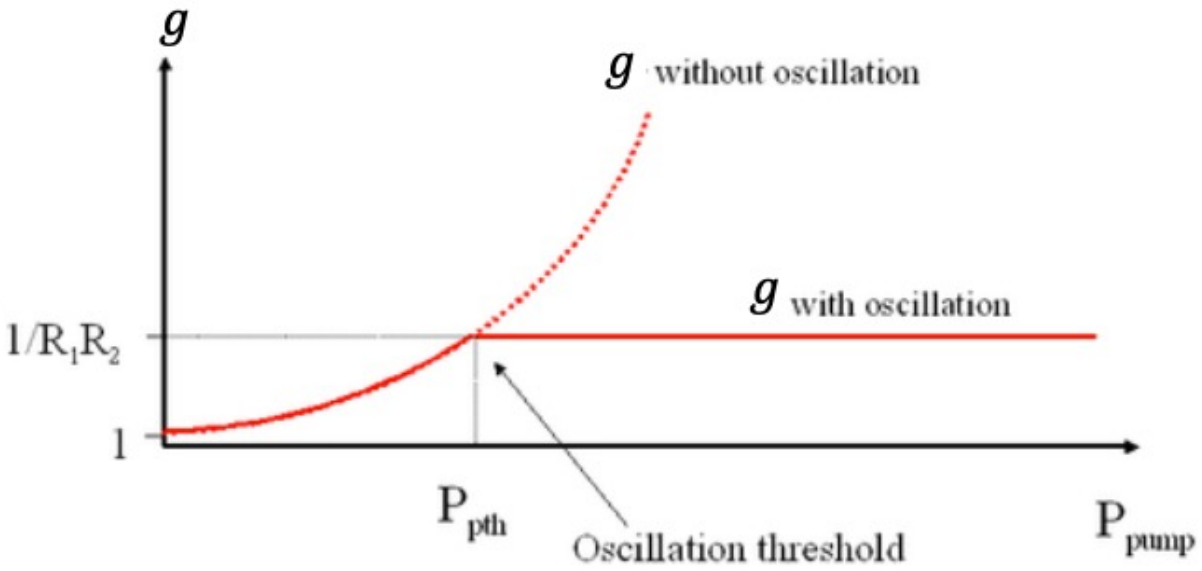
Gain-loss Profile

The gain function is described in the figure below. It represents the rate of increase of the number of photons inside the cavity due to stimulated emission.



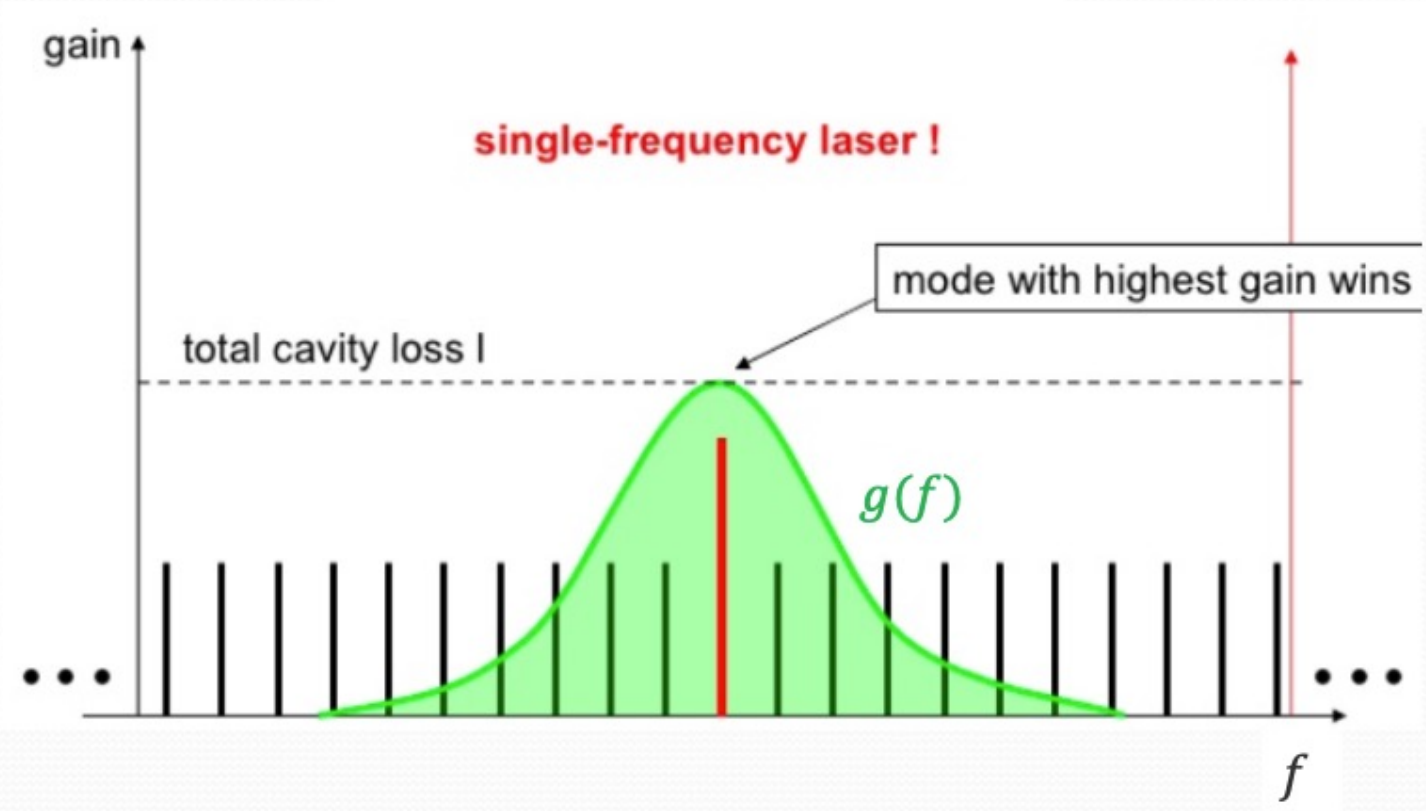
Gain-loss Profile

The gain function is described in the figure below.



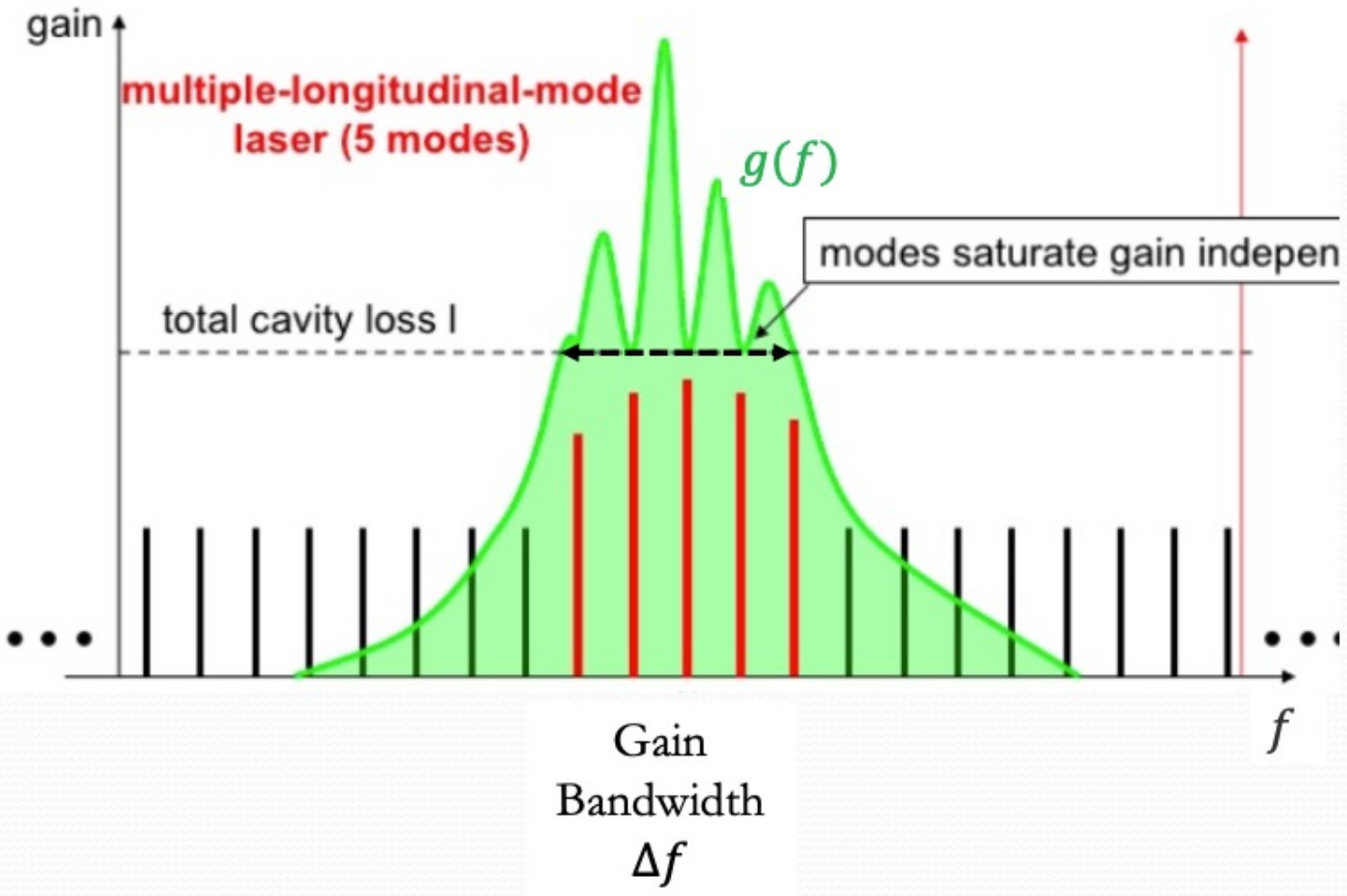
Lasing Spectrum

Laser single-mode operation.



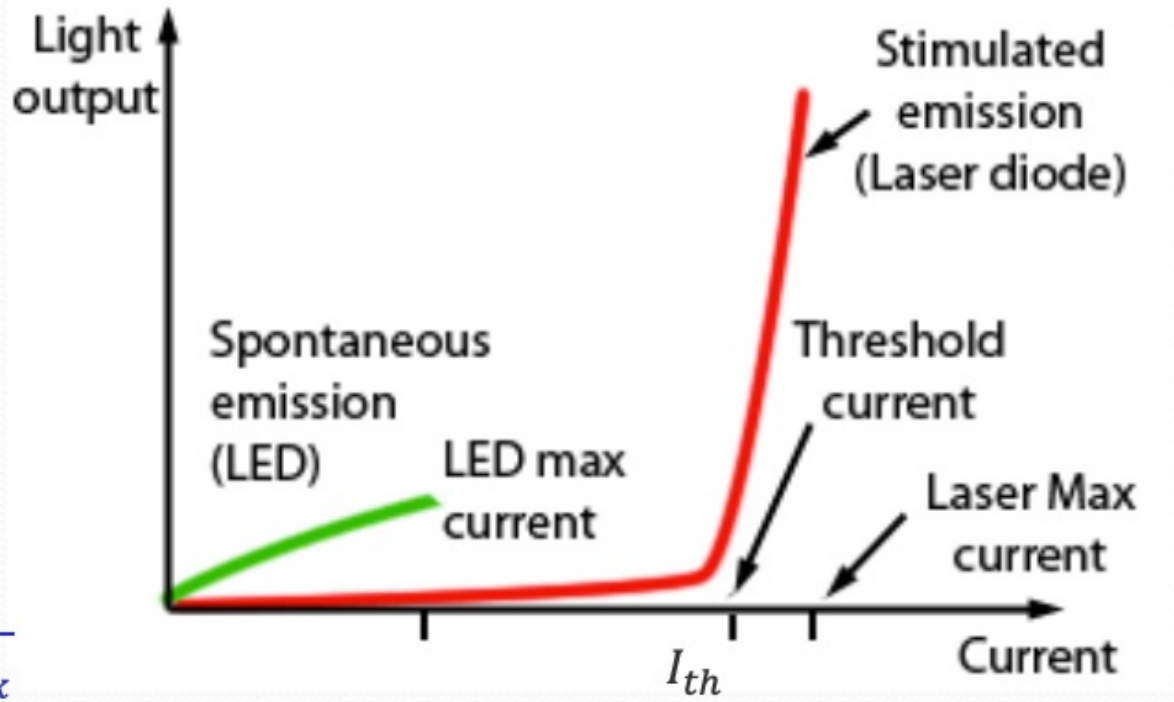
Lasing Spectrum

Laser multi-mode operation.



Output Optical Power

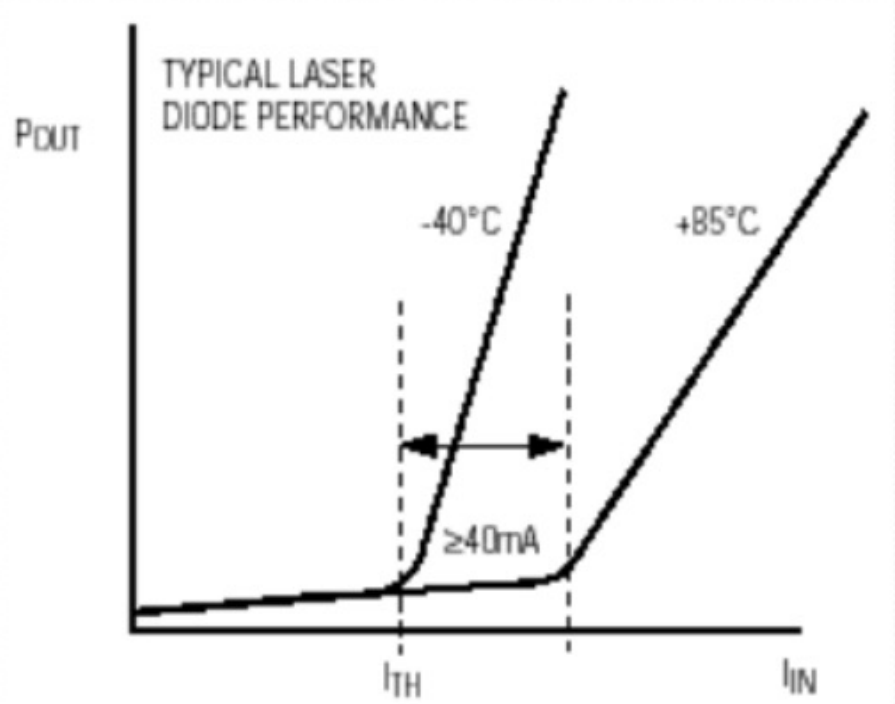
PI curve



$$I_{th} \sim \frac{1}{\lambda_{peak}^2}$$

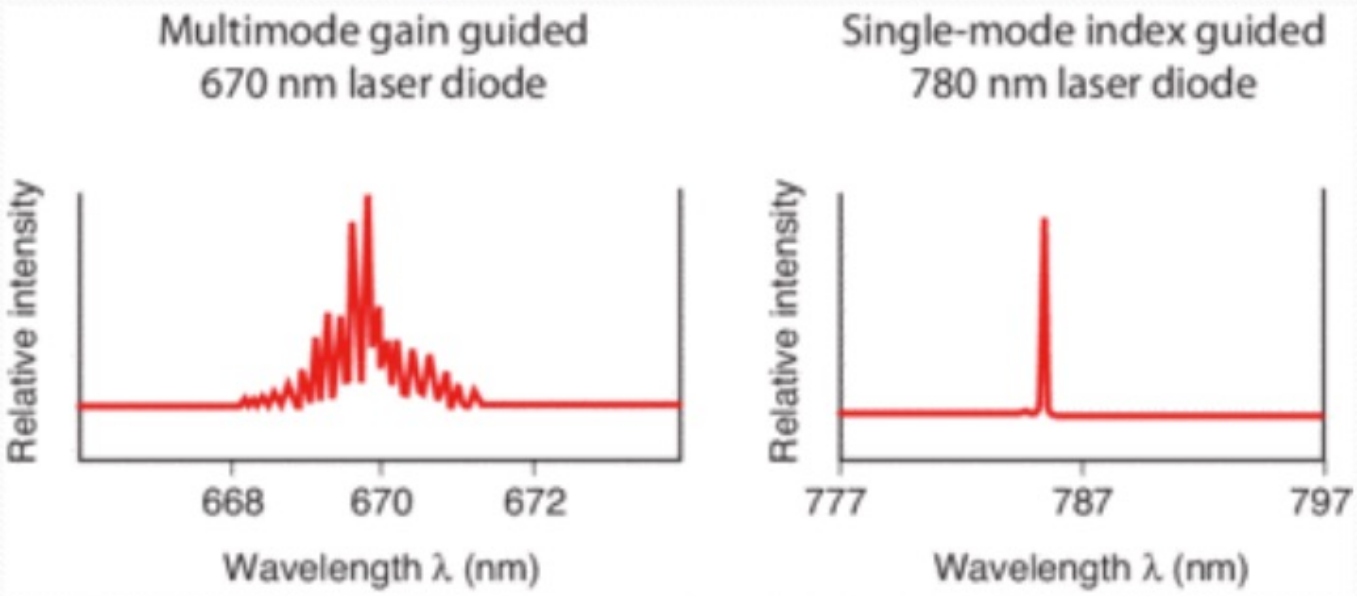
Output Optical Power

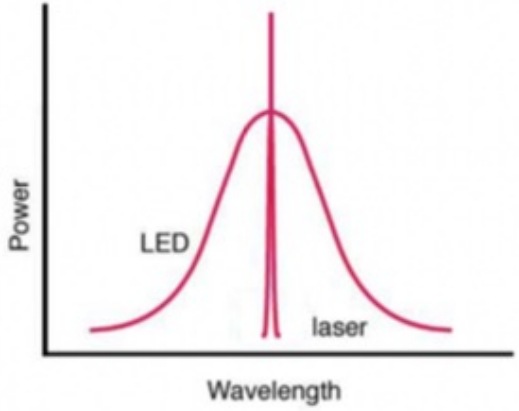
PI curve



Lasing Spectrum

Single and multi-mode laser spectrum





Example. Consider an AlGaAs (n=3.7) laser diode emitting at peak wavelength of 870 nm. The cavity length is 200 μm and the gain spectrum has a bandwidth of 6 nm. Find the number of modes with gain.

Mode index of peak wavelength $\lambda_m = \frac{2nL}{m} \Rightarrow m = \frac{2nL}{\lambda_m} = 1701$

Mode separation around peak wavelength $\delta\lambda_m = \frac{2nL}{m^2} = 0.5115$ nm

Number of modes with gain $\frac{\Delta\lambda}{\delta\lambda} = \frac{6}{0.5115} = 11$

For single-mode operation $L = 20\ \mu m \Rightarrow \delta\lambda_m = 5.115$ nm

$$\frac{\Delta\lambda}{\delta\lambda} = \frac{6}{5.115} = 1$$

Semiconductor Lasers Features

1) Temporal coherence, due to very small linewidth

2) Spatial coherence, producing collimated beams

3) Diffraction-limited and hence very focused beams.

Q & A

Send me an email if you have any questions.